Problem 17.28 (modified)

$V_{DD} =$

An inverter has a 3.3V supply. It draws a static current of 60 µA in one state and 0 µA in the other. When the inverter is switched at 100 Mhz, the inverter draws 150 µA. Estimate the load capacitance.

$C?$ $\sqrt{f}$

$$P_{total} = P_{dyn} + P_{static}$$

$$P_{static} = \frac{V_{DD}\, i_{cycle}}{2}$$

$$P_{total} = V_{DD} i_{tot} = (3.3V)(150\mu A) = P_{dyn} + 3.3V(30\mu A)$$

$$P_{dyn} = (3.3V)(150\mu A - 30\mu A) = 396\mu W$$

$$= C V_{DD}^2 f$$

$$C = \frac{P_{dyn}}{V_{DD}^2 f} = \frac{396\mu W}{(3.3V)^2\, 100\times10^6} = 3.6\times10^{-13} F$$

$$364\ fF$$

Exercise 17.13

An inverter is scaled from 0.5 µm to a 0.13 µm technology, and V_{DD} is scaled from 5V to 1.2V. By what factor does P_{dyn} decrease? Note: C scales linearly with feature size.

$$\frac{C_{0.5}}{C_{0.13}} = \frac{0.5}{0.13}$$

$$\frac{P_{dyn\,0.5}}{P_{dyn\,0.13}} = \frac{C_{0.5}\,V_{DD\,0.5}^{2}\,\cancel{f}}{C_{0.13}\,V_{DD\,0.13}^{2}\,\cancel{f}} = \left(\frac{0.5}{0.13}\right)\frac{(5V)^2}{(1.3V)^2} = 66.8$$

Problem 17.26

An IC inverter fabricated in a 65-nm CMOS process is found to have a load capacitance of 3 fF. If the inverter is operated from a 1.2-V power supply,

a) Find the energy needed to charge and discharge the load capacitance.

b) If the IC chip has 5 million of these inverters operating at an average switching frequency of 2.5 GHz, what is the power dissipated in the chip?

c) What is the average current drawn from the power supply?

$E = CV_{dd}^2$

$= 3fF(1.2V)^2$

$E = 4.32fJ$

$P_1 = f(CV_{dd}^2)$ ← E

$= 2.5\times10^9(4.32fJ) = 10.8\mu W$

$P_T = (5\times10^6)(10.8\mu W) = 54W$

$P = IV$

$54W = (1.2V)I$, $I = \frac{54W}{1.2V} = 45A$